CLASS 6 — MATHS · FORMULA SHEET

Chapter 1: **Patterns in Mathematics** | Quick revision, one page

Number patterns

- Sequence: a list of numbers following a rule; the numbers are called terms.
- Common difference: the same number added each time, e.g. $5, 9, 13, 17, \ldots$ (add 4).
- Counting numbers: $1, 2, 3, 4, \ldots$; odd numbers: $1, 3, 5, 7, \ldots$; even numbers: $2, 4, 6, 8, \ldots$
- The nth odd number $= 2n - 1$; the nth even number $= 2n$.

Special number families

- Square numbers: $1, 4, 9, 16, 25, \ldots$ ($n \times n$).
- Cube numbers: $1, 8, 27, 64, 125, \ldots$ ($n \times n \times n$).
- Triangular numbers: $1, 3, 6, 10, 15, 21, \ldots$ (add $1, 2, 3, 4, \ldots$ in turn).
- Virahanka (Fibonacci) numbers: $1, 2, 3, 5, 8, 13, \ldots$ (each term = sum of the two before).

Useful results

- Sum of the first n odd numbers $= n \times n$.
- Sum of the first n natural numbers $= \dfrac{n(n+1)}{2}$ (the nth triangular number).
- Two consecutive triangular numbers add up to a square number, e.g. $6 + 10 = 16$.
- Complete graph on n points: number of lines $= \dfrac{n(n-1)}{2}$.

Steps to explore a pattern

- Write the first few terms; find the difference between neighbours; state the rule; test the rule on the next term.

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